CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 8: Logarithms

Topic-wise short notes with solved examples

The meaning of a logarithm as an index, changing between exponential and logarithmic forms, the three laws of logarithms and their use in simplifying and solving.

1 Definition

- If $a^x = N$, where $a > 0$, $a \neq 1$ and $N > 0$, then x is the **logarithm of N to the base a**.

$$a^x = N \quad \text{means the same as} \quad \log_a N = x$$

- A logarithm is an **index**: $\log_a N$ is the power to which a must be raised to get N.
- Logarithms are defined only for **positive** numbers. The base is positive and not 1.

Exponential form	Logarithmic form
$2^5 = 32$	$\log_2 32 = 5$
$10^3 = 1000$	$\log_{10} 1000 = 3$
$3^{-2} = \frac{1}{9}$	$\log_3 \frac{1}{9} = -2$
$9^{\frac{1}{2}} = 3$	$\log_9 3 = \frac{1}{2}$

- Logarithms to base 10 are **common logarithms**. When no base is written, base 10 is understood.

2 Basic results

- $\log_a 1 = 0$, because $a^0 = 1$.
- $\log_a a = 1$, because $a^1 = a$.
- $\log_a a^n = n$ and $a^{\log_a N} = N$.
- $\log_{10} 10 = 1$, $\log_{10} 100 = 2$, $\log_{10} 0.1 = -1$, $\log_{10} 0.01 = -2$.

Example 2.1 Find $\log_2 64$, $\log_5 \frac{1}{125}$ and $\log_{\sqrt{3}} 9$.

SOLUTION

$2^6 = 64; 5^{-3} = \frac{1}{125}; (\sqrt{3})^4 = 9$

6, -3 and 4

Example 2.2 Find x if $\log_x 81 = 4$, and if $\log_3 x = -2$.

SOLUTION

$x^4 = 81 = 3^4$, so $x = 3$. And $x = 3^{-2}$.

$x = 3; x = \frac{1}{9}$

Example 2.3 Find x if $\log_8 x = \frac{2}{3}$.

SOLUTION

$x = 8^{\frac{2}{3}} = (2^3)^{\frac{2}{3}}$

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3 Laws of logarithms

- All logarithms in a law must have the same base.

Law	Statement
Product	$\log_a(mn) = \log_a m + \log_a n$
Quotient	$\log_a \frac{m}{n} = \log_a m - \log_a n$
Power	$\log_a m^n = n \log_a m$

- They follow from the laws of indices: if $m = a^x$ and $n = a^y$ then $mn = a^{x+y}$.
- There is **no** law for $\log(m+n)$ or $\log(m-n)$.

Example 3.1 Express $\log \frac{a^2 b}{\sqrt{c}}$ in terms of $\log a$, $\log b$ and $\log c$.

SOLUTION

$\log a^2 + \log b - \log c^{\frac{1}{2}}$

$2 \log a + \log b - \frac{1}{2} \log c$

Example 3.2 Write as a single logarithm: $2 \log 3 + \log 5 - \log 9$.

SOLUTION

$\log 9 + \log 5 - \log 9$

$\log 5$

Example 3.3 Evaluate $\log 25 + \log 4$ and $\log_2 96 - \log_2 3$.

SOLUTION

$\log 100 = 2$ and $\log_2 32 = 5$

2 and 5

Example 3.4 Simplify $\dfrac{\log 27}{\log 9}$.

SOLUTION

$\dfrac{3\log 3}{2\log 3}$

$\dfrac{3}{2}$

4 Using given values

- Break each number into factors whose logarithms are known.
- $\log 5$ can be found from $\log 2$: $\log 5 = \log \dfrac{10}{2} = 1 - \log 2$.

Example 4.1 Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 12$, $\log 5$ and $\log 1.5$.

SOLUTION

$\log 12 = 2\log 2 + \log 3 = 0.6020 + 0.4771$

$\log 5 = 1 - 0.3010$

$\log 1.5 = \log 3 - \log 2$

1.0791, 0.6990 and 0.1761

Example 4.2 If $\log 2 = x$ and $\log 3 = y$, express $\log 72$ and $\log 4.5$ in terms of x and y.

SOLUTION

$72 = 2^3 \times 3^2$ and $4.5 = \dfrac{9}{2}$

$3x + 2y$ and $2y - x$

5 Equations with logarithms

- Combine the logarithms on each side into one, using the laws.
- If $\log_a m = \log_a n$, then $m = n$.
- Change a number into a logarithm when needed: $1 = \log 10$, $2 = \log 100$.
- **Reject** any value that makes the number inside a logarithm zero or negative.

Example 5.1 Solve $\log(x+3)+\log(x-3)=\log 16$.

SOLUTION

$\log(x^2-9)=\log 16$, so $x^2=25$ and $x=\pm 5$.

$x=-5$ makes $x+3$ negative.

$x=5$

Example 5.2 Solve $\log x+\log 5=2$.

SOLUTION

$\log 5x=\log 100$

$x=20$

Example 5.3 Solve $2\log x-\log(x+6)=0$.

SOLUTION

$\log\dfrac{x^2}{x+6}=\log 1$, so $x^2-x-6=0$ and $x=3$ or $x=-2$.

$x=3$

Example 5.4 If $\log(a+b)=\log a+\log b$, express a in terms of b.

SOLUTION

$a+b=ab$, so $a(b-1)=b$

$a=\dfrac{b}{b-1}$

Example 5.5 If $a^2+b^2=23ab$, show that $\log\dfrac{a+b}{5}=\dfrac{1}{2}(\log a+\log b)$.

SOLUTION

$(a+b)^2=25ab$, so $\left(\dfrac{a+b}{5}\right)^2=ab$.

Taking logarithms: $2\log\dfrac{a+b}{5}=\log a+\log b$

6 Common mistakes

- Writing $\log(m+n) = \log m + \log n$ or $\log(m-n) = \log m - \log n$.
- Writing $\dfrac{\log m}{\log n} = \log m - \log n$. The quotient law is for $\log \dfrac{m}{n}$.
- Writing $(\log m)^n = n \log m$. The power law is for $\log m^n$.
- Keeping a negative solution that makes a logarithm undefined.
- Forgetting that $\log 1 = 0$, not 1.
- Mixing bases inside one law.

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